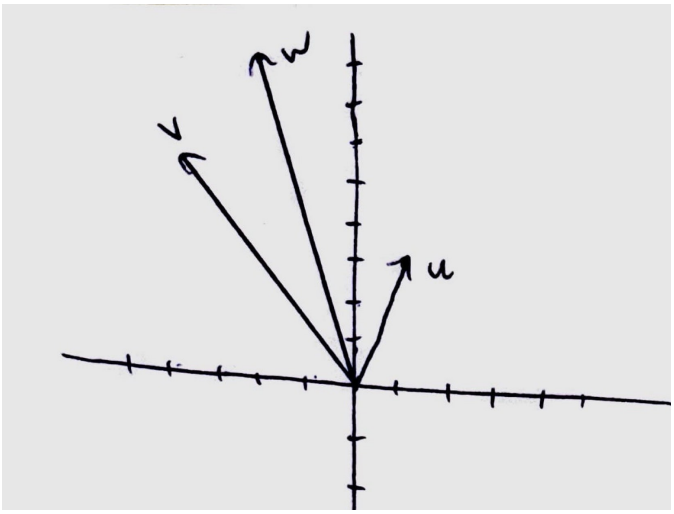
Multivariable Calculus Summer Assignment

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Problem 1

(ab)



(c) $\|u\| = \sqrt{(1)^2 + (3)^2} = \sqrt{1+9} = \boxed{\sqrt{10}}$

$\|v\| = \sqrt{(-4)^2 + (5)^2} = \sqrt{16+25} = \boxed{\sqrt{41}}$

$\|w\| = \|\langle -3, 8\rangle\| = \sqrt{(-3)^2 + (8)^2} = \sqrt{9+64} = \boxed{\sqrt{73}}$

(d) $u + v = \langle 1, 3\rangle + \langle -4, 5\rangle = \boxed{\langle -3, 8\rangle}$

$\|u+v\| = \|w\| = \boxed{\sqrt{73}}$

(e) Vectors a and b must be orthogonal, which is exactly when $a^T b = b^T a = a \cdot b = 0$.

Proof. Let a and b be vectors such that $\|a\| + \|b\| = \|a+b\|$. By definition, this is equivalent to:

$$\begin{aligned} a^T a + b^T b &= (a+b)^T (a+b) \\ &= \left(a^T + b^T\right)(a+b) \\ \cancel{a^T a} + \cancel{b^T b} &= \cancel{a^T a} + a^T b + b^T a + \cancel{b^T b} \\ 0 &= a^T b + b^T a \end{aligned}$$

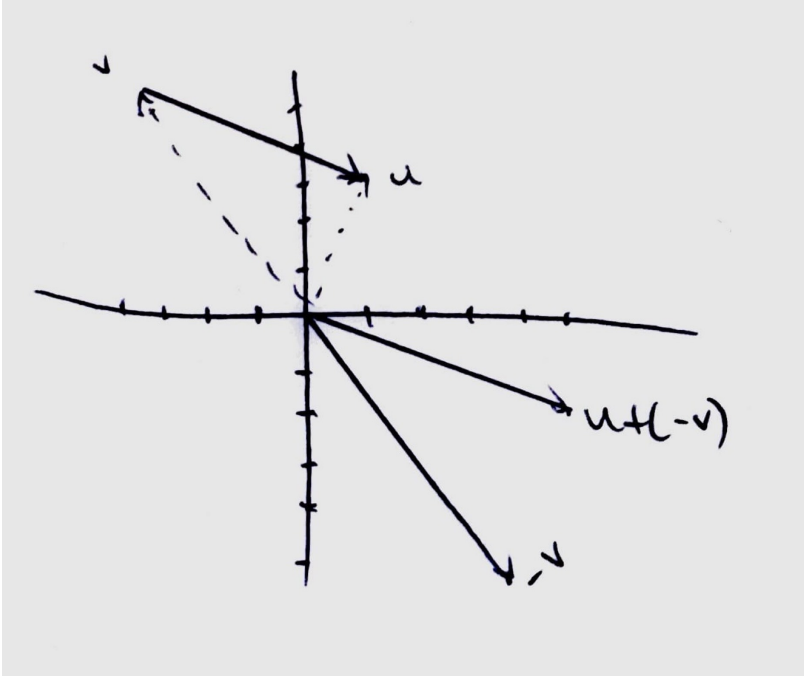
The dot product is commutative: $a \cdot b = b \cdot a$ so $a^T b = b^T a$.

Thus we must have $2a^T b = 0 \iff \boxed{a^T b = a \cdot b = 0 \iff a \text{ and } b \text{ are orthogonal}}$ for $\|a\| + \|b\| = \|a+b\|$, which is the equality case of the triangle inequality/Pythagorean theorem. □

Problem 2

(abc) $-v = -\langle -4, 5\rangle = \boxed{\langle 4, -5\rangle}$

$u + (-v) = \langle 1, 3\rangle + \langle 4, -5\rangle = \boxed{\langle 5, -2\rangle}$



(c) This vector is the same as the vector found in part (b).

Problem 3

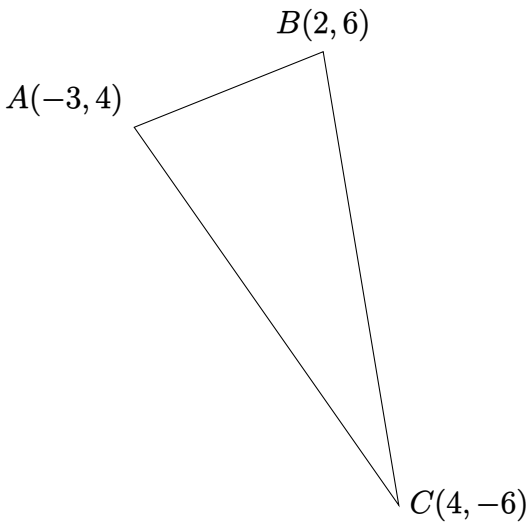
(a) AB is equivalent to CD.

(b) BC is equivalent to AD.

(c) $AM = \dfrac{AC}{2} = \dfrac{AB + BC}{2}$.

(d) $AB + BC - AC = AC - AC = 0$.

Problem 4



(a) $AB = B - A = \langle 2-(-3), 6-4\rangle = \boxed{\langle 5,2\rangle}$

$BC = C - B = \langle 4-2, -6-6\rangle = \boxed{\langle 2,-12\rangle}$

$CA = A - C = \langle -3-4, 4-(-6)\rangle = \boxed{\langle -7,10\rangle}$

(b) $AB + BC + CA = \boxed{\langle 0,0\rangle}$

(c) $$\frac{1}{2}AC - \frac{1}{2}AB = \frac{1}{2}CA - \frac{1}{2}AB$$
$$= -\frac{1}{2}\langle -7,10\rangle - \frac{1}{2}\langle 5,2\rangle = \left\langle \frac{7}{2}, -5\right\rangle - \left\langle \frac{5}{2}, 1\right\rangle$$
$$= \left\langle \frac{7}{2} - \frac{5}{2}, -5-1\right\rangle = \boxed{\langle 1,-6\rangle} = \frac{1}{2}BC$$

(d) $$AC = C - A = \langle 4,-6,5\rangle - \langle -3,4,2\rangle = \langle 7,-10,3\rangle$$
$$AB = B - A = \langle 2,6,-1\rangle - \langle -3,4,2\rangle = \langle 5,2,-3\rangle$$
$$BC = C - B = \langle 4,-6,5\rangle - \langle 2,6,-1\rangle = \langle 2,-12,6\rangle$$
$$\frac{1}{2}AC - \frac{1}{2}AB = \frac{1}{2}\langle 7,-10,3\rangle - \frac{1}{2}\langle 5,2,-3\rangle = \left\langle \frac{7}{2}, -5, \frac{3}{2}\right\rangle - \left\langle \frac{5}{2}, 1, -\frac{3}{2}\right\rangle$$
$$= \left\langle \frac{7}{2} - \frac{5}{2}, -5-1, \frac{3}{2} + \frac{3}{2}\right\rangle = \boxed{\langle 1,-6,3\rangle} = \frac{1}{2}\langle 2,-12,6\rangle = \frac{1}{2}BC$$

$AC/2 - AB/2$ is still a scalar multiple of BC.

Problem 5

(a) $\boxed{n = \langle s, -r\rangle}$.

(b) $m \cdot n = rs + s(-r) = rs - rs = \boxed{0}$.

(c) $(2n)\cdot(-3m) = \langle 2s,-2r\rangle \cdot \langle -3r,-3s\rangle = (2s)(-3r) + (-2r)(-3s) = -6sr + 6rs = 0$. Geometrically, this means $2n$ and $-3m$ are also perpendicular.

(d) $u \cdot v = \langle 1,3\rangle \cdot \langle -4,5\rangle = (1)(-4) + (3)(5) = -4 + 15 = 11$ so u and v are not perpendicular.

(e) The dot product is commutative.

Proof. Let $a = (a_1, a_2, \ldots, a_n)$ and $b = (b_1, b_2, \ldots, b_n)$ be vectors in $\mathbb{R}^n$, and $a_i, b_i \in \mathbb{R}$. Then:

$$a \cdot b = a_1b_1 + a_2b_2 + \ldots + a_nb_n = \sum_{k=1}^{n} a_kb_k$$

$$b \cdot a = b_1a_1 + b_2a_2 + \ldots + b_na_n = \sum_{k=1}^{n} b_ka_k = \sum_{k=1}^{n} a_kb_k$$

Scalar multiplication a_kb_k is commutative. Since $a \cdot b = b \cdot a$, the dot product is commutative. $\square$

Proof. Alternatively, let a, b be vectors in $\mathbb{R}^n$. The dot product is defined as:

$$a \cdot b = \|a\| \, \|b\| \cos\theta \tag{1}$$

$$b \cdot a = \|b\| \, \|a\| \cos\theta \tag{2}$$

where θ is the angle between vectors a and b. Since vector norms $\|a\|$ and $\|b\|$ are scalar values, and scalar multiplication is commutative, we have $\|a\|\|b\| = \|b\|\|a\|$. Since θ remains the same in (1) and (2), as the angle between vectors do not change, $a \cdot b$ must equal $b \cdot a$. □

Problem 6

(a) DeGray's displacement vector $\boxed{v = \langle 10, 30, 15 \rangle}$. The distance to his starting point is $\|v\| = \sqrt{(10)^2 + (30)^2 + (15)^2} = \sqrt{100 + 900 + 225} = \sqrt{1225} = 35$ meters.

(b) Seltzer's displacement vector $\boxed{w = \langle 40, -60, -10 \rangle}$. The straight line distance from DeGray to Seltzer is $\boxed{w - v} = \langle 30, -90, -25 \rangle$.

(c) $D = \left\| \left\langle 50 \cdot 10^6, 20 \cdot 10^6, 30 \cdot 10^6, 80 \cdot 10^6 \right\rangle \right\| = \sqrt{\left(50 \cdot 10^6\right)^2 + \left(20 \cdot 10^6\right)^2 + \left(30 \cdot 10^6\right)^2 + \left(80 \cdot 10^6\right)^2}$

$= \sqrt{10^{12}\left(50^2 + 20^2 + 30^2 + 80^2\right)} = \sqrt{10^{12}}\sqrt{2500 + 400 + 900 + 6400}$

$= 10^6\sqrt{100\left(25 + 4 + 9 + 64\right)} = 10^6\sqrt{100}\sqrt{102} = 10^7\sqrt{102}$

The ship needs to travel $\boxed{10^7\sqrt{102} \text{ meters}}$ to return to the sun.

Problem 7

(a) The vertices of the cube are: $(0,0,0)\,, (2,0,0)\,, (0,2,0)\,, (0,0,2)\,, (2,2,0)\,, (2,0,2)\,, (0,2,2)\,, (2,2,2)$. Also, $A = (2,1,0)\,, B = (2,2,1)\,, C = (1,2,2)$.

(b) $OA = \langle 2,1,0 \rangle\,, OB = \langle 2,2,1 \rangle\,, OC = \langle 1,2,2 \rangle$.

(c) $BA = \boxed{OA - OB} = \langle 0,-1,-1 \rangle$ and $BC = \boxed{OC - OB} = \langle -1,0,1 \rangle$.

(d) $BA \cdot BC = \langle 0,-1,-1 \rangle \cdot \langle -1,0,1 \rangle = (0)(-1) + (-1)(0) + (-1)(1) = \boxed{-1}$. This means BA and BC are not perpendicular.

(e) $\|OA\| = \sqrt{(2)^2 + (1)^2 + (0)^2} = \sqrt{4 + 1 + 0} = \boxed{\sqrt{5}}$

$\|OB\| = \sqrt{(2)^2 + (2)^2 + (1)^2} = \sqrt{4 + 4 + 1} = \sqrt{9} = \boxed{3}$

$\|OC\| = \sqrt{(1)^2 + (2)^2 + (2)^2} = \sqrt{1 + 4 + 4} = \sqrt{9} = \boxed{3}$

$\|BA\| = \sqrt{(0)^2 + (-1)^2 + (-1)^2} = \sqrt{0 + 1 + 1} = \boxed{\sqrt{2}}$

$\|BC\| = \sqrt{(-1)^2 + (0)^2 + (1)^2} = \sqrt{1 + 0 + 1} = \boxed{\sqrt{2}}$

(f) $BA - BC = \langle 0,-1,-1 \rangle - \langle -1,0,1 \rangle = \boxed{\langle 1,-1,-2 \rangle}$

$\|BA - BC\| = \|\langle 1,-1,-2 \rangle\| = \sqrt{(1)^2 + (-1)^2 + (-2)^2} = \sqrt{1 + 1 + 4} = \boxed{\sqrt{6}}$

(g) By the definition of the dot product, we have $BA \cdot BC = \|BA\|\|BC\| \cos\theta$, where θ is the angle between vectors BA and BC, which equals $\angle ABC$.

$$BA \cdot BC = \|BA\|\|BC\| \cos\theta$$

$$\implies -1 = \sqrt{2}\sqrt{2}\cos\theta \implies -\frac{1}{2} = \cos\theta \implies \theta = \arccos\left(-\frac{1}{2}\right) = \boxed{\frac{2\pi}{3}}$$

Problem 8

$$r(t) = \langle 2\cos t, 2\sin t \rangle$$

(a) By the Pythagorean identity $\cos^2\theta + \sin^2\theta = 1$:

$$\|r(t)\| = \sqrt{(2\cos t)^2 + (2\sin t)^2} = \sqrt{4\cos^2 t + 4\sin^2 t} = \sqrt{4}\sqrt{\cos^2 t + \sin^2 t} = 2\sqrt{1} = \boxed{2}$$

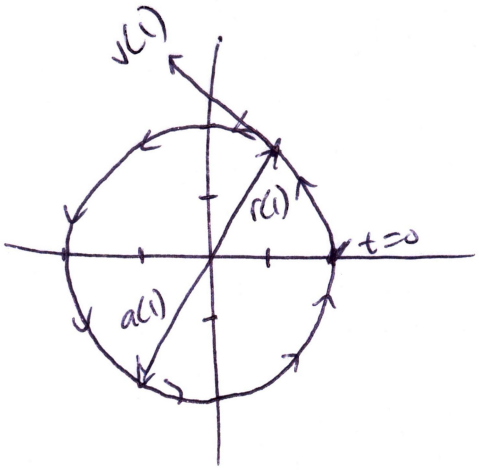
(b) $v(t) = r'(t) = \left\langle \frac{\mathrm{d}}{\mathrm{d}t} 2\cos t, \frac{\mathrm{d}}{\mathrm{d}t} 2\sin t \right\rangle = \boxed{\langle -2\sin t, 2\cos t \rangle}$.

(c) $v(t) \cdot r(t) = \langle 2\cos t, 2\sin t \rangle \cdot \langle -2\sin t, 2\cos t \rangle = -4\cos t \sin t + 4\sin t \cos t = 0$.

(d) $a(t) = v'(t) = \left\langle \frac{\mathrm{d}}{\mathrm{d}t}\left[-2\sin t\right], \frac{\mathrm{d}}{\mathrm{d}t}\left[2\cos t\right] \right\rangle = \boxed{\langle -2\cos t, -2\sin t \rangle}$.

(e) $a(t) = \langle -2\cos t, -2\sin t \rangle = -1\,\langle 2\cos t, 2\sin t \rangle = -1 r(t)$. $k = -1$.

(f)



Problem 9

(a) $u \cdot v = (1)(2) + (2)(10) + (-3)(5) = 2 + 20 - 15 = \boxed{7}$.

(b)

$$\underbrace{\begin{bmatrix} 1 & 2 & -3 \\ 2 & 0 & -2 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} 2 & 0 & 1 \\ 10 & -2 & 3 \\ 5 & 5 & 2 \end{bmatrix}}_{B} = \underbrace{\boxed{\begin{bmatrix} 7 & -19 & 1 \\ -6 & -10 & -2 \end{bmatrix}}}_{AB}$$

(c) The first column and first row of the result of matrix multiplication is 7, which is the dot product between vectors u and v. This is because u is the first row of the left hand side matrix A, and v is the first column of right hand side matrix B.

In general, for any matrix product AB, we have:

$$(AB)_{ij} = (\text{row } i \text{ of A}) \cdot (\text{column } j \text{ of B})$$

where $(AB)_{ij}$ is the entry at the ith row and jth column of C.